20/3/26

Day - 5.

1) ~~Distance~~ = 60 km/h.
speed

$$\frac{60}{18} \times \frac{5}{18} \Rightarrow 16.666\ m/s$$

16.6..
3 | 50
30
20
18
20
18
2

16.67 m/s

2)

$\because$ Average speed = $\frac{\text{Total Distance}}{\text{Total time}}$

Distance $\Rightarrow$ $40 \times 2 = 80$ km
D = S×T
$60 \times 2 = 120$ km

Total Distance = 200 km
Total time = 2+2 $\Rightarrow$ 4 hrs

Average speed = $\frac{200}{4}$
= 50 km/hr

3)
$\because$ Relative speed = a+b (opposite direction)
= 50 + 70
= 120 km/hr

4) $\because$ Time = $\frac{\text{Distance}}{\text{speed}}$

$\frac{15}{2}$ = 7.5 km (each)

walking $\Rightarrow$ 7.5/5 = 1.5 hrs

Running $\Rightarrow$ 7.5/10 = 0.75 hrs

Total = 1.5 + 0.75 $\Rightarrow$ 2.25 hrs

$\therefore$ 2 hours 15 minutes

5)

length (Distance) = 100m

time = 10 seconds

$\frac{100}{10} = 10 m/s$

to km/h :

$10 \times \frac{18}{5} = 36 km/h$

∴ 10m/s or 36km/h

Block 2

Verbal Ability

Reading Passage :

1) Algorithms are important because it to optimize system performance

2) If inefficient algorithms are used Programs run slower.

3) optimization means improving a program or algorithm so that it gives better performance

Grammar

1) He have been working here since 2 years

He has been working here for 2 years

2) She did not wrote the code

She did not write the code

3) They is preparing for interviews

They are preparing for interviews

Block 3:

Artificial Intelligence.

Artificial Intelligence is a technology that enables machines to think and learn like humans. It helps systems analyze data, recognize patterns and make decisions.

For example, virtual assistants like Siri and Google assistant use AI to respond to voice commands.

In real life, AI is used in healthcare for diagnosis, in banking for fraud detection and in automation

systems.

overall, AI makes technology smarter, faster and more efficient